

NAVNEET SINGH ARORA

SENIOR AI ENGINEER · AGENTIC SYSTEMS & RAG

 Hamburg, Germany |  |  |  |  | 

PROFILE

Senior AI Engineer with **8+ years of experience** building production agentic systems and RAG pipelines. Currently leading AI at **Repath** — architecting multi-agent climate resilience solutions with **LangGraph**, **LangChain**, **FastMCP**, and **RAGAS/DeepEval** evaluation on Google Cloud. Track record spans AI Lead ownership (0→1), team leadership at NICE and Cognizant, measurable delivery (saved \$40K, cut deployment time 95%), and published open-source work (3+ n8n workflows on the official gallery, vectools npm package). Core expertise: agentic system design, RAG architecture, vector search (Qdrant), and LLM evaluation — delivered to production, not just prototyped.

WORK EXPERIENCE

Repath

Dec 2021 – Present
Hamburg, Germany

Machine Learning Engineer (AI Lead)

- **Led the AI engineering function** (built from 0→1 as the first AI hire), owning end-to-end technical strategy through production deployment on Google Cloud Platform.
- Designed and built a **production multi-agent AI system** using **LangGraph** and **LangChain**, orchestrating LLM workflows across heterogeneous climate data sources; evaluated with **RAGAS** and **DeepEval** for hallucination control and retrieval quality.
- Built an **MCP integration layer** using FastMCP, exposing data analysis tools as agent-callable functions, enabling internal stakeholders to query and explore climate datasets without engineering support.
- Implemented **enterprise-grade security** across the AI stack: OAuth 2.0 authentication, end-to-end encryption, and containerised deployment with Docker.
- Led data preparation strategy and CI/CD pipeline for ML integration workflows.
- Designed and built the **API layer** using **FastAPI** and **Prisma** (TypeScript), structuring ML model outputs into typed REST endpoints consumed by the platform frontend.
- Managed **~100TB of climate and geospatial data** across the AI pipeline.

Omdena

Jun 2021 – Dec 2021
Remote

Chapter Lead

- **Led the Hamburg Chapter** as community lead, coordinating collaborative AI projects across distributed contributors.
- Hosted and drove open-source AI projects to facilitate knowledge-sharing and applied ML practice.

NICE

Jan 2018 – Oct 2020
Pune, India

Tech Lead

- **Led a cross-functional team** of developers and QA engineers in software analysis and platform rebuilding.
- Delivered full-stack features using **AngularJS** (frontend) and **Java with Spring framework** (backend).
- Designed and developed **Tableau dashboards** with parameter-driven performance tuning.

Cognizant

Jul 2015 – Jan 2018
Pune, India

Program Analyst → Project Innovation Lead

- Promoted to Project Innovation Lead within the **Credit Suisse engagement** for consistently driving technical improvements.
- Saved **\$40,000 in client costs** (Credit Suisse) through process optimisation and system redesign.
- Automated the deployment system, reducing deployment time from 4 hours to 20 minutes — a **95% reduction**.
- Backend development using Java and Shell Scripts.

ADDITIONAL EXPERIENCE

Student Tutor, Computer Vision — Universität Hamburg · Hamburg, Germany · Oct 2021 – Mar 2022

Full Stack Engineer (Freelance) — Allooffice.space · Remote · May 2020 – Feb 2021

PROJECTS

Agent-Forge

2026
Ongoing

FileAgent — Autonomous File Organiser

- Building an LLM-driven file organiser agent in Python using **LangGraph**, with a scanner and executor in place and a Plan-then-Apply safety pattern in progress.
- Packaged inside Agent-Forge: a multi-agent monorepo with shared execution cores and structured JSON logging.

RAG Workbench

2026
Ongoing

RAG-Lab — Hybrid RAG Research Workbench

- Architecting a **5-engine RAG benchmark** (Naive, Hybrid, Parent-Child, Agentic, Vectorless) with a unified interface for side-by-side retrieval comparison using **Qdrant**.
- Built an evaluation suite with **RAGAS** and **DeepEval** — curating 30-item golden datasets (faithfulness, context precision, relevancy) to quantify hallucination and retrieval quality across strategies.

Librarian Wiki

2026
Ongoing

Wiki Librarian — Agentic Knowledge OS

- Engineered a Claude-powered Obsidian vault agent with custom slash commands (/ingest, /evolve, /meditate) for autonomous knowledge management.
- Built with **Claude Code**; implements multi-agent orchestration, **MCP server integration**, and decision logging.

Open Source

2026
Ongoing

vectools & n8n Automation Workflows

- **vectools (npm)**: Published ESM-native TypeScript library for vector mathematics and AI embeddings.
- **n8n Workflows**: Published **3+ workflows** on official gallery, covering Gemini-based parser and Notion integrations.

RESEARCH & ACADEMIC

MSc Thesis

Universität Hamburg
2023

Audio-Visual Emotion Recognition in Continuous Domain

- Multi-modal valence-arousal prediction using dual-stream self-supervised learning: contrastive audio + FaceSynthetics visual models. Automated pseudo-labelling pipeline, evaluated on IEMOCAP with PyTorch.

Collaborations

Universität Hamburg
2022

COLMAN & IndoRE

- **COLMAN**: Embodied AI object-goal navigation with CLIP semantic embeddings and Transformer architectures.
- **IndoRE**: Relation Extraction using mBERT and RoBERTa for low-resource Indian languages (Augmentation & distant supervision).

Adaptive Gaze

Universität Hamburg
2021

FAZE: Few-Shot Adaptive Gaze Estimation

- DT-ED network with MAML meta-learning for gaze estimation personalised from 3–9 calibration samples.

EDUCATION

Universität Hamburg

2020 – 2023

MSc, Intelligent Adaptive Systems

Hamburg, Germany · Thesis: Audio-Visual Emotion Recognition

Lovely Professional University

2011 – 2015

BSc, Computer Science & Engineering

Punjab, India

SKILLS

TECHNICAL EXPERTISE

- **AI / ML**: LLMs, Fine-tuning, Multi-Agent Systems, RAG Pipelines, Vector Search, Evals (RAGAS, DeepEval), FastMCP, LangChain, LangGraph, PyTorch, Qdrant
- **Languages**: Python, Java, JavaScript, TypeScript, Shell Scripting, SQL
- **Frontend/Backend**: FastAPI, REST APIs, Prisma, Tkinter, AngularJS, Spring Framework
- **Infrastructure**: GCP, Docker, OAuth 2.0, E2E Encryption, CI/CD, Linux, MySQL
- **Tools**: Claude Code, Codex, Gemini, Cursor, Ollama, n8n, Tableau, Git, Jira, Comet ML

LANGUAGES

- **English**: Fluent (C1)
- **German**: Elementary (A2)
- **Hindi & Punjabi**: Native